E-Commerce Recommendation System

Using Data Mining, Graph Analysis, and BERT-Based Sentiment Analysis

A University Project Report

Datasets: Instacart Market Basket | Amazon Fine Food Reviews

1. Introduction

Online retail has become one of the most data-rich domains in modern commerce. With millions of transactions happening every day, there is a tremendous opportunity to use that data to understand customers better and make smarter recommendations. This project sets out to build an intelligent recommendation system for e-commerce by combining three complementary techniques: association rule mining, graph-based analysis, and deep learning-based sentiment analysis.

The core idea is that purchasing behavior data and customer reviews together tell a more complete story than either source alone. By mining co-purchase patterns, we can identify which products are frequently bought together. By modeling those relationships as a network, we can measure which products are most central and influential. And by analyzing the language customers use in their reviews, we can gauge whether the overall experience with a product is positive, negative, or neutral.

The project is split into two major tasks. The first task focuses on the Instacart dataset, a collection of real grocery transactions, and applies FP-Growth association rule mining followed by PageRank-based graph analysis. The second task uses the Amazon Fine Food Reviews dataset to fine-tune a BERT language model for three-class sentiment classification. The combination of these techniques lays the foundation for a smarter, more context-aware recommendation engine.

2. Data Collection

Two publicly available datasets were used throughout this project, each chosen for its relevance to a specific part of the pipeline.

2.1 Instacart Market Basket Dataset

The Instacart dataset captures real-world grocery shopping behavior. It contains millions of order records, each describing which products a customer purchased during a single trip. The dataset includes product names, department and aisle classifications, and order-level metadata such as the time of day and day of the week the purchase was made. For this project, only the product and order data were needed, since the goal was to discover which items are frequently purchased together.

2.2 Amazon Fine Food Reviews Dataset

The Amazon Fine Food Reviews dataset contains over half a million reviews of food products sold on Amazon. Each record includes the review text, a summary headline, a numeric rating from one to five stars, and various product and user identifiers. This dataset was ideal for sentiment analysis because the star ratings provide a natural ground truth for labeling reviews as positive, negative, or neutral. The richness of the review text also makes it well-suited for a language model like BERT, which benefits from long, expressive input.

3. Data Preprocessing

3.1 Transaction Data (Instacart)

The full Instacart dataset is large enough to cause memory and runtime issues on standard hardware, so a representative sample of 50,000 orders was selected for analysis. This reduction maintained diversity while keeping computation feasible. The preprocessing steps applied were:

- Product names were merged with order data to give each transaction human-readable labels.
- Orders were grouped into baskets, where each basket is the list of products purchased together in a single order.
- Products appearing in fewer than 200 orders were removed to reduce noise and focus on items with sufficient statistical signal.

- Baskets with fewer than two items were discarded, since single-item baskets cannot form meaningful association rules.
- One-hot encoding was applied to the basket data to prepare it for the FP-Growth algorithm.

These steps reduced the dataset to a clean and computationally efficient form without sacrificing the patterns that matter most.

3.2 Text Data (Amazon Reviews)

Preparing the review data for sentiment analysis involved both cleaning and restructuring. First, columns not relevant to sentiment — such as user IDs, product IDs, and timestamps — were dropped. The review summary and full text were then concatenated into a single input field, which gives the model more context than either piece alone.

Star ratings were mapped to sentiment labels: ratings of 1 or 2 were labeled Negative, a rating of 3 was labeled Neutral, and ratings of 4 or 5 were labeled Positive. This mapping is intuitive and aligns with how most people interpret star ratings in practice.

One important challenge was class imbalance. As shown in Figure 1 below, the original dataset is dominated by positive reviews, with negative and neutral reviews being much less common. Training a model on imbalanced data tends to produce biased results, where the model simply learns to predict the majority class. To address this, the dataset was balanced by randomly sampling 10,000 examples from each class, resulting in 30,000 total training examples with equal representation.

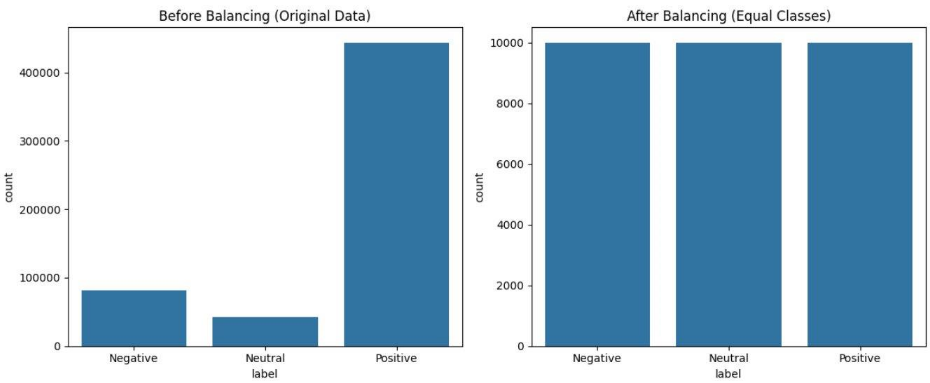

Figure 1: Sentiment distribution before balancing (left) and after balancing (right). The original data is heavily skewed toward positive reviews. After undersampling, each class contains exactly 10,000 examples.

4. Association Rule Mining

Association rule mining is a technique used to discover interesting relationships between variables in large datasets. In the context of market basket analysis, it helps identify which products tend to be purchased together. For this project, the FP-Growth (Frequent Pattern Growth) algorithm was used because it is significantly more efficient than the classic Apriori algorithm, especially for large transaction datasets.

4.1 Key Metrics

Three standard metrics were used to evaluate the strength of discovered rules:

- Support: The fraction of all baskets that contain a given itemset. Higher support means the combination appears frequently.
- Confidence: Given that product A is in a basket, the probability that product B is also present. High confidence indicates a strong directional relationship.
- Lift: The ratio of observed co-occurrence to what would be expected if the items were chosen independently. A lift above 1.0 means the products appear together more often than chance, indicating a genuine association.

4.2 Parameters and Results

The algorithm was run with a minimum support threshold of 0.02 and a maximum itemset length of 3. These parameters were chosen to focus on meaningful, recurring patterns while avoiding the combinatorial explosion of very long itemsets.

The scatter plot in Figure 2 shows the relationship between support and confidence for all discovered rules, with bubble size proportional to lift. The plot reveals that the strongest rules (highest lift, shown in darker red) tend to cluster at higher confidence values, even when support is relatively low. This is typical in grocery data, where niche but coherent shopping habits appear in a small fraction of baskets but remain highly predictive.

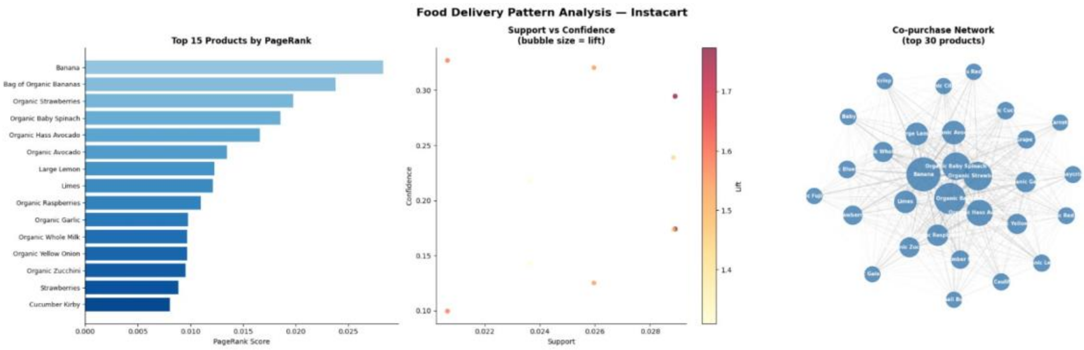

Figure 2: Left — Top 15 products by PageRank score. Center — Support vs. Confidence scatter plot with bubble size proportional to lift value. Right — Co-purchase network graph showing the top 30 products and their relationships.

The strongest association rule discovered was:

Organic Hass Avocado → Bag of Organic Bananas (Lift ≈ 1.77)

This rule means that customers who purchase organic avocados are approximately 77% more likely to also purchase organic bananas than a random customer. This kind of association likely reflects a broader pattern of health-conscious shopping: customers buying organic produce tend to buy multiple organic items in the same trip. From a business perspective, this is a clear opportunity for product bundling and co-placement strategies.

5. Graph-Based Analysis (PageRank)

Association rules tell us which pairs or groups of products are connected, but they do not directly tell us which individual products are the most influential in the overall co-purchase network. To answer that question, a graph-based approach was used.

5.1 Network Construction

A co-purchase network was built where each node represents a product, and an edge between two nodes indicates that those products have been purchased together at least once in the sampled data. Edge weight can reflect the frequency of co-purchase, making the graph a natural representation of the overall purchasing ecosystem.

5.2 PageRank Analysis

The PageRank algorithm, originally developed for ranking web pages, was applied to this product network. In the web context, a page is considered important if many other important pages link to it. Applied here, a product receives a high PageRank score if many other popular products are frequently co-purchased with it. In other words, PageRank measures how central a product is in the overall shopping network.

The bar chart in Figure 2 (left panel) shows the top 15 products by PageRank score. Banana topped the ranking by a clear margin, followed by Bag of Organic Bananas. This is not surprising — bananas are a staple grocery item that appears in many different shopping contexts, making them a hub in the co-purchase network. Their high centrality means they act as connectors, bridging diverse shopping behaviors.

The network graph in Figure 2 (right panel) gives a visual sense of this structure. Products like Organic Baby Spinach, Banana, and Organic Strawberries sit near the center, densely connected to many others. Peripheral nodes, by contrast, tend to be more specialized items with fewer co-purchase links.

5.3 Business Implication

PageRank scores can directly inform recommendation system design. Products with high centrality are natural candidates for homepage features, bundle deals, and cross-category promotions. Knowing that bananas connect a wide range of shopping trips means they can be used as an anchor product — if a customer buys bananas, it is safe to infer they are open to buying many of the products around bananas in the network.

6. BERT-Based Sentiment Analysis

While transaction data tells us what people buy, review text tells us how they feel about what they bought. Sentiment analysis of product reviews is a valuable signal for any recommendation system: recommending products with a history of negative reviews would undermine user trust. This section describes how a BERT-based model was trained to classify reviews into three sentiment categories.

6.1 Model Architecture

BERT (Bidirectional Encoder Representations from Transformers) is a pre-trained language model developed by Google that understands language in both left-to-right and right-to-left directions simultaneously. This bidirectional context makes it particularly powerful for classification tasks because it captures nuances that simpler models miss.

For this project, the bert-base-uncased variant was used as the base model. A classification head with three output neurons was added on top of the pre-trained encoder to distinguish between Negative, Neutral, and Positive sentiment. The model was fine-tuned on the balanced 30,000-sample dataset using the Hugging Face Transformers library, which provides an accessible and well-tested implementation of BERT in Python.

6.2 Training Configuration

The review summary and body text were concatenated and tokenized using BERT's built-in tokenizer, which handles subword tokenization and special tokens like [CLS] and [SEP] automatically. The model was trained for a small number of epochs given the computational cost of fine-tuning a transformer, and the balanced dataset ensured that no single class dominated the gradient updates.

6.3 Performance Results

The model achieved an overall accuracy of approximately 83% on the held-out test set. The confusion matrix in Figure 3 provides a breakdown of predictions across the three classes, while the classification metrics chart in Figure 4 shows precision, recall, and F1-score for each class individually.

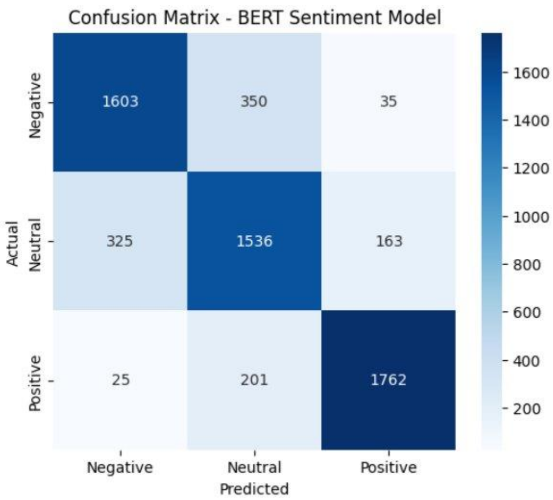

Figure 3: Confusion matrix for the BERT sentiment classifier. Rows represent actual labels; columns represent predicted labels. The model correctly identifies the majority of samples in all three classes.

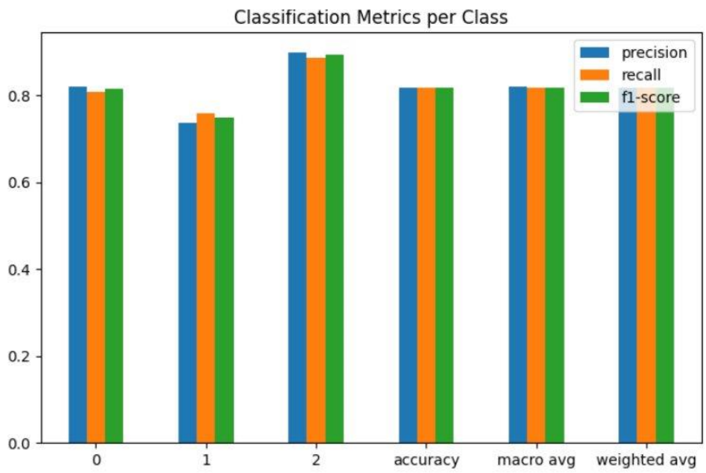

Figure 4: Classification metrics (precision, recall, and F1-score) broken down by class. Class 0 = Negative, Class 1 = Neutral, Class 2 = Positive.

Looking at the confusion matrix, the model correctly classified 1,603 of the negative reviews, 1,536 of the neutral ones, and 1,762 of the positive ones. The most common error was misclassifying neutral reviews — 325 were predicted as negative and 163 as positive. This pattern is expected: neutral reviews often contain a mix of positive and negative language, making them genuinely ambiguous even for human readers. The model performed best on the positive class, which likely benefits from more distinct vocabulary and a clearer emotional tone.

From the metrics chart, the positive class (class 2) achieved the highest scores across precision, recall, and F1, all close to 0.89. The negative class (class 0) performed at around 0.81 across the board. The neutral class (class 1) was the weakest, with recall around 0.75, consistent with the confusion matrix findings. The macro-average F1 of approximately 0.82 reflects solid overall performance across all three classes.

7. Visualizations Summary

Visualization played an important role in communicating the findings of this project clearly. Each chart was designed to highlight a specific aspect of the analysis, and together they tell the full story from raw data to actionable insight.

- Figure 1 (Sentiment Distribution): Demonstrates the dramatic class imbalance in the original Amazon dataset and confirms that the balancing step successfully equalized the three sentiment classes. Without this intervention, the model would have been biased toward predicting Positive.
- Figure 2 – Left (PageRank Bar Chart): Ranks the top 15 products by their importance in the co-purchase network. The dominance of Banana and Bag of Organic Bananas highlights their role as shopping-basket anchors.
- Figure 2 – Center (Support vs Confidence Scatter Plot): Maps out the space of discovered association rules. The color gradient representing lift makes it easy to spot rules that are not only frequent but genuinely associative rather than coincidental.
- Figure 2 – Right (Co-purchase Network Graph): Provides an intuitive spatial representation of product relationships. Central nodes are visually larger and more connected, making the influence hierarchy immediately apparent.
- Figure 3 (Confusion Matrix): Breaks down model predictions at the sample level. The near-diagonal concentration of values confirms that the model is reliable, while off-diagonal entries highlight where errors occur and why.
- Figure 4 (Classification Metrics): Summarizes model performance numerically across all classes. The consistently high bars for the positive class and somewhat shorter bars for the neutral class reinforce the interpretation from the confusion matrix.

8. Business Insights and Recommendations

The results of this project have direct implications for how e-commerce platforms can make smarter decisions. The following insights emerge from the combined analysis:

8.1 Product Bundling and Cross-Selling

The association rule between Organic Hass Avocado and Bag of Organic Bananas, with a lift of approximately 1.77, represents a clear opportunity for cross-selling. When a customer adds avocados to their cart, the system should proactively suggest organic bananas. More broadly, any rule with a lift above 1.5 indicates a pairing strong enough to drive meaningful uplift in average order value.

8.2 Category-Level Targeting

The strong co-purchase patterns observed among organic products suggest a coherent segment of health-conscious shoppers. These customers respond well to organic labels, premium pricing, and fresh produce categories. E-commerce platforms can use this insight to build tailored recommendation modules for this segment rather than relying on generic suggestions.

8.3 PageRank-Driven Homepage and Promotions

Products with high PageRank scores, particularly Banana and Organic Baby Spinach, are well-positioned for homepage features, daily deals, and email promotions. Because these products connect many parts of the shopping network, promoting them tends to activate a wide range of downstream purchases. This makes them high-leverage items for marketing investment.

8.4 Sentiment-Aware Filtering

The BERT model's 83% accuracy means it can reliably flag products with predominantly negative or neutral reviews. A recommendation system can use this signal to downrank or exclude poorly reviewed products from suggestions, improving customer satisfaction and reducing returns. It can also highlight consistently positive products as "highly rated" to build trust.

8.5 Complementary Strengths

Perhaps the most important insight is that neither the transaction data nor the review data alone is sufficient. Association rules find patterns but cannot evaluate quality. Sentiment analysis evaluates quality but cannot suggest what to pair with what. Together, they enable a recommendation system that is both relevant (based on purchasing behavior) and trustworthy (based on review sentiment).

9. Conclusion

This project demonstrated that combining data mining, graph analysis, and deep learning can produce meaningful and actionable insights for e-commerce recommendation systems. Each technique contributed something distinct: FP-Growth uncovered co-purchase patterns, PageRank identified the most influential products in the shopping network, and fine-tuned BERT captured the sentiment embedded in customer language.

The strongest findings were the avocado-banana association rule (lift of approximately 1.77), the dominant centrality of Banana in the PageRank ranking, and the BERT model's 83% accuracy on a balanced three-class sentiment task. The model's performance was solid overall, with the main challenge being the classification of neutral reviews, which are inherently ambiguous and sit on the boundary between positive and negative.

From a technical standpoint, several deliberate choices improved the quality of results: sampling 50,000 orders kept the transaction analysis tractable without losing representativeness; balancing the review dataset prevented class-bias in model training; and using the combined summary-and-text field gave BERT richer context for classification.

Looking ahead, several extensions could make this system even more powerful. Training on the full review dataset rather than 10,000 samples per class would likely improve accuracy further. Incorporating temporal data from the Instacart orders could reveal seasonal purchasing trends. And deploying the recommendation pipeline in a real-time API would allow the system to generate suggestions on the fly as customers browse.

Overall, this project showed that the combination of structured transactional data and unstructured textual data, analyzed with the right tools, can form the backbone of a practical, intelligent recommendation system that serves both business goals and customer needs.

— End of Report —